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yellow = must know · **pink** = trigger words in the question

CH 3 Option strategies

Rate type · multiply or divide?

```
"effective six-month rate" → ×(1+r) once
"continuously compounded"   → e^(rT)
to expiry (FV): ×(1+r)   ×e^(+rT)
to today  (PV): ÷(1+r)   ×e^(−rT)
```

Premiums are paid today → grow them. (F − K) happens later → discount it.

Positions & payoffs at expiry

POSITION	MEANS	PAYOFF
Long call	right to buy at K	`max(S_T−K,0)`
Short call	must sell at K	`−max(S_T−K,0)`
Long put	right to sell at K	`max(K−S_T,0)`
Short put	must buy at K	`−max(K−S_T,0)`
Long fwd	must buy at F	`S_T − F`
Short fwd	must sell at F	`F − S_T`

S_T = price on expiry day. Long = bought, short = sold. Call: S first. Put: K first.

Profit

```
profit = payoff − FV(net premium paid)
FV = premium × (1 + r)  or  × e^(rT)
```

LOOK FOR **"payoff"** = no premiums. **"profit"**, **"opportunity cost"**, **"carrying the premium forward"** = grow & subtract.

Put-call parity

```
C − P = PV(F − K) = (F − K)/(1 + r)
stock: C − P = S₀ − PV(K)
```

LOOK FOR **"verify parity"** → compute both sides · **"fair price of the put"** → P = C − (F − K)/(1+r) · **"same premium"** → only when K = F

Parity arbitrage

LOOK FOR **"quotes violate parity"**, **"riskless profit"**

1. Fair C − P vs quoted C − P.
2. Quoted too high → **sell call + buy put** (sells at K) + **long forward** (buys at F).
3. Profit at T = (K − F) − FV(net premium paid). S_T must cancel.

Synthetic forward

```
long call(K) + short put(K) = long fwd at K
payoff = S_T − K
profit = S_T − K − FV(C − P)
```

Strike vs premium

```
K ↑ → call cheaper, put pricier
K ↓ → call pricier, put cheaper
```

LOOK FOR **"zero-cost collar"**: call premium = put premium. Need more from the call → move call strike **down**.

Collar on a held asset (buy put K_1, sell call K_2)

```
value at T: K₁ if S_T<K₁ | S_T | K₂ if S_T>K₂
net prem = put − call  (+ pay, − receive)
min profit = K₁ − (S₀ + net)(1 + r)
max profit = K₂ − (S₀ + net)(1 + r)
```

LOOK FOR **"total initial outlay"** = S_0 + net premium · **"piecewise"** = the 3 zones

Asymmetric butterfly ($K_1 < K_2 < K_3$)

LOOK FOR **"write n of the middle calls"**, **"payoff zero above K_3"**

```
λ = (K₃ − K₂)/(K₃ − K₁)
buy λn at K₁, (1 − λ)n at K₃
max payoff (at K₂) = λn·(K₂ − K₁)
max profit = max payoff − FV(net prem)
```

```
or: x + y = n ;  x·K₁ + y·K₃ = n·K₂
net prem = x·C₁ + y·C₃ − n·C₂ (debit)
low BE:  x(S_T − K₁) = FV(prem)
high BE: x(S_T−K₁) − n(S_T−K₂) = FV(prem)
max loss = −FV(prem)
```

Check λ: $\lambda K_1 + (1-\lambda)K_3 = K_2$. More calls on the side closer to K_2.

Zone method (build any payoff)

Strikes split S_T into zones. In each zone a call is "on" if S_T > K, a put if S_T < K. Add the "on" pieces → piecewise payoff. Test numbers only to check.

Other spreads

Bull: buy low-K call, sell high-K call. **Bear**: reverse. **Straddle**: long call + put, same K. **Strangle**: put K < call K.

CH 4 Risk management

Natural positions

Producer = naturally **long** → short forward, buy put (floor), or collar. **Buyer** = naturally **short** → long forward, buy call (cap).

Producer profit per unit

LOOK FOR **"fixed cost"** + **"variable cost"** = total cost · **"evaluate at $X"**

```
unhedged:  S_T − cost
short fwd: F − cost          (flat)
long put:  max(S_T, K) − cost − FV(P)
collar:    [Kp | S_T | Kc] − cost
           − FV(P − C)
```

Floor = profit when S_T ≤ K_put. **Cap** = profit when S_T ≥ K_call. Net premium < 0 → you receive → raises profit.

Produce or shut down?

LOOK FOR **"should it still produce?"**

Fixed cost is sunk. Produce if the price > **variable** cost, even if total profit is negative.

Why hedging adds value

Concave profits → hedging raises expected value. Causes: taxes, bankruptcy/distress costs, costly external financing, debt capacity, managerial risk aversion.

```
no refund:  ½·G(1−t) + ½·(−L)
hedged:     E[pre-tax]·(1 − t)
full credit: loss × (1−t) → no gain
carry-fwd:  refund t·L ÷ (1 + r)
```

Not to hedge: transaction costs, expertise, monitoring, tax/accounting, collateral.

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BA II Plus: e^x = [2nd][LN] · ln = [LN] · [STO]/[RCL] keeps full precision

CH 5 Forwards & futures

Step 0 · every question

1. **T in years**: days ÷ 365 · months ÷ 12. Store it.
2. % → decimal: 2.40% = 0.024.
3. Type rT first, then [2nd][LN]. **"do not round intermediate"** → [STO]/[RCL].

Which formula? trigger words

QUESTION SAYS...	USE
"non-dividend-paying"	S_0e^{rT}
"dividend yield", index	$S_0e^{(r-\delta)T}$
"pays a dividend of $"	$(S_0-\Sigma PV\ D)e^{rT}$
"prepaid forward"	$S_0 - PV(div)$
"you are long/short", **"delivery price"**, **"present value of the contract"**	$(F-K)e^{-rT}\times N$
"quoted at", **"riskless profit"**	fair F vs quote
"bid/ask", "cost k", 2 rates	bounds F^+ F^-
"implied" yield	$r - \ln(F/S)/T$
"exchange rate"	$x_0e^{(r-r_f)T}$
"margin", **"mark-to-market"**	margin balance

Forward price

```
no divs:  F = S₀·e^(rT)
yield δ:  F = S₀·e^((r − δ)T)
$ divs:   F = (S₀ − ΣDᵢe^(−rtᵢ))·e^(rT)
```

No dividends → F > S_0 always. Only dividends paid **before delivery** count. Quarterly t = 0.25, 0.5, 0.75, 1.

Prepaid forward (pay today, receive later)

```
no divs: F^P = S₀
yield δ: F^P = S₀·e^(−δT)
$ divs:  F^P = S₀ − ΣDᵢe^(−rtᵢ)
then     F = F^P·e^(rT)
```

Value of an existing forward

```
LONG  = (F − K)·e^(−rT) × shares
SHORT = (K − F)·e^(−rT) × shares
```

1. Fair F = S_0e^{rT} for the **remaining** T.
2. Long wins if F > K; short wins if K > F.
3. **Discount** with e^{-rT} (most-missed step).
4. × shares. Answer format: **"$F; $value"**.

Why: exit with the opposite forward at today's F → left with F − K at maturity. New contract: K = F → value 0.

Arbitrage (quoted F ≠ fair F)

```
quote > fair → CASH-AND-CARRY
  borrow, buy stock, short forward
  profit = F_quote − S₀e^(rT)
quote < fair → REVERSE CASH-AND-CARRY
  short stock, lend, long forward
  profit = S₀e^(rT) − F_quote
```

Sell the expensive one, buy the cheap one. Profit is at expiration.

Bounds with transaction costs

```
F⁺ = (S_ask + 2k)·e^(r_borrow·T)
F⁻ = (S_bid − 2k)·e^(r_lend·T)
```

Above F^+ → cash-and-carry. Below F^- → reverse. Between → no arbitrage.

Implied yield · premium · currency

```
δ = r − ln(F/S₀)/T
fwd premium = F/S₀; annual = ln(F/S₀)/T
currency: F = x₀·e^((r_dom − r_for)T)
```

Foreign rate acts like a dividend yield. r_dom > r_for → F > spot.

Synthetics

```
forward = stock − bond
stock   = forward + bond
bond    = stock − forward
```

Futures & margin

```
notional = N × 250 × F
new bal = old bal·e^(rΔt) + N×250×ΔF
```

Long gains when F rises. **Initial margin**: posted to open. **Maintenance** ≈ 70–80% of initial; below → **margin call** back to initial.

Futures vs forwards

Futures: exchange-traded, standardized, clearinghouse is counterparty, **marked to market daily**, low credit risk, easy to close. **Forwards**: OTC, custom, settled at maturity. Prices ≈ equal for short contracts. Uses: asset allocation, cross-hedging, lower costs.

Practice set, worked

```
Q2 212.50·e^(.024×107/365) = 214.00
Q3 87.25·e^(.031×355/365)   = 89.92
Q4 146.80·e^(.0185×276/365) = 148.87
Q5 F = 101.8777
   5.4777 × 0.993838 × 2,500 = 13,610
Q6 F = 70.8399 (short)
   2.6601 × 0.989555 × 4,000 = 10,529
Q1 λ=.6 → 6×1400, 4×1650; prem 133.154
   600 − 133.154×1.02 = 464.18
```

The wrong answers are mistakes

F below S_0 (no divs)	used e^{-rT}
= today's price	forgot to grow
way too high	T = 1, not days/365
value a bit too big	forgot e^{-rT}
right size, minus sign	long/short flipped
uses $S_0 as F	skipped fair F
butterfly 600 / 466.85	no premium / not grown

Final check

Payoff or profit? · Paid = subtract, received = add · Long/short sign? · Effective or continuous? · If your answer is one step short, it's the trap.